

enterprise-okf-ai Handbook

Operational and technical manual for onboarding, running, validating, and maintaining this repository.

Start Here

Who this handbook is for

- AI/ML engineers implementing or extending the pipeline

- con

Role-based navigation

- New contributor path: [Tutorial lane](#tutorial-lane-first-successful-run) -> [Reference lane](#reference-lane-lookup-details)

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Tutorial Lane (first successful run)

Goal: run the project end-to-end and produce validated artifacts.

Step 1: install dependencies

```
uv sync --extra dev --frozen
```

```
cp .e
```

Step 2: run quality gates

```
UV_CACHE_DIR=.uv-cache make check
```

Expected result:

- ruff

Step 3: run strict end-to-end pipeline

```
UV_CACHE_DIR=.uv-cache make run-e2e
```

Expected summary output:

- `val

Artifacts produced:

- `arti

How-to Lane (common operational tasks)

Build an OKF bundle from source docs

```
uv run enterprise-okf-ai build-okf examples/enterprise_docs artifacts/local_okf_bundle
```

Validate bundle integrity

```
uv run enterprise-okf-ai okf-validate --okf-dir artifacts/local_okf_bundle
```

Build graph artifacts

```
uv run enterprise-okf-ai graph-build --okf-dir artifacts/local_okf_bundle
```

Run retrieval query with traces

```
uv run enterprise-okf-ai retrieve-search "Which API updates order status?" --with-trace
```

Run agentic Q&A

```
uv run enterprise-okf-ai agent-ask "Which API updates order status and what does it depend on?"
```

Run agent benchmark evaluation

```
uv run enterprise-okf-ai agent-eval \
```

```
--b
```

Start runtime services

```
make run-api
```

```
make
```

Generate handbook PDF

```
make handbook-pdf
```

Reference Lane (lookup details)

Glossary

- OKF-style bundle: markdown concept files with YAML frontmatter.

- Con

Core frontmatter contract

Required keys in each concept document:

- `id`

Configuration reference

Runtime settings are loaded through ``enterprise_okf_ai.core.settings.Settings``.

Important settings:

- ``okf``

Config files:

- ``conf``

Module map

Canonical package: ``src/enterprise_okf_ai/``

- ``ingestion``: multi-format parsing + normalization

- ``okf``

Compatibility package:

- ``src``

Output artifacts and interpretation

Primary outputs from strict E2E:

- ``arti``

Key current metrics (from latest strict run):

- ``grap``

Community and repository health files

- CONTRIBUTING.md

Explanation Lane (why the system is designed this way)

Why OKF-style markdown

- portable and versionable in git

Why strict validation before retrieval/agent

- prevents broken links from degrading graph and retrieval quality

Why hybrid retrieval instead of vector-only

- vector retrieval captures semantic similarity

Why explicit unsupported-answer logic

- reduces confident hallucinations in weak-evidence scenarios

Troubleshooting

CLI command not found

If `enterprise-okf-ai` is unavailable in shell path:

```
PYTHONPATH=src uv run --no-sync python -m enterprise_okf_ai.cli.main --help
```

Failing E2E due to stale artifacts

Delete target artifact folder and rerun strict pipeline:

```
rm -rf artifacts/e2e_real_run
```

GitHub CLI authentication issues

```
gh auth status
```

Non-fatal dependency warning in tests

Current tests may show a `chromadb` deprecation warning (`asyncio.iscoroutinefunction`). The run is still suc

cessful.

Documentation Maintenance Policy

- Treat `README.md` as onboarding + quick-run surface.

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Appendix: Environment-specific notes from this session

This environment originally exposed a non-functional mounted `.git` path. A local git-dir workaround (`.git-local`) was used to run git operations safely in-session. This is not a project runtime requirement and should not be treated as standard setup for normal clones.

Official References and Sources

Project references:

- Goo

Official docs for stack used here:

- uv: l

Documentation quality guidance used in this handbook rewrite:

- GitH

